

Spherical Bodies: World Shape Lab

GROW • Science • 40 minutes

I can collect fair comparison observations and combine them with real space evidence to describe the Sun, Earth and Moon as approximately spherical.

Prepare before pupils arrive

Setup: Mark one viewing distance and three viewpoints. Prepare a large ball, a flat disc and local real-evidence cards showing spacecraft views, an eclipse shadow and curved day-night edges.

Safety: Avoid small counters where ingestion is a risk, keep models on the work surface and follow school rules for any photographs.

Equipment: Large ball, rigid card disc, three-position viewing mat, six-row record and pencil or camera on a school device.

Safety: Use large clean models, keep them on the table, maintain clear walkways and use a camera only under school routines.

Fallback: Use the prepared model and real-space card set; record similarities, differences and the conclusion on paper.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Describe the Sun, Earth and Moon as approximately spherical bodies.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Last lesson: why was one circular view insufficient? Last lesson: what happened to the disc outline as it turned?	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	observation — what is noticed comparison — same test for both conclusion — what the evidence means Place the disc and sphere at the same marked distance. Observe both from the front without naming the expected answer.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Turn the six model observations into a comparison: the disc outline changes; the sphere's outline stays round. Add real observations: spacecraft views keep showing round whole-body outlines, Earth's eclipse shadow is curved, and the curved day-night edge moves across Earth and the Moon.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Sort the model results and real space observations, then freeze enough evidence for a careful conclusion. Choose one recorded observation.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Complete the fair three-view comparison and present an evidence-led conclusion about Solar System body shape. Record: Six model observations, one named real space observation, an approximately spherical conclusion and one model limitation.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Cite one paired model view and one real space observation to justify approximately spherical. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Use the same distance and the same three viewpoints for both.

Correct. Matching distance and viewpoints makes outline differences meaningful.

Misconception repair

Repaired. Approximately spherical describes the overall shape without erasing mountains, valleys, craters or slight flattening.

Guided checkpoint

Correct. Matching distance and viewpoints makes outline differences meaningful. A fair comparison changes the object while keeping the viewing method the same.

Front result

Disc record. Disc record: its flat face narrows as the viewpoint changes. A sphere would retain a round outline after this turn.

Quarter-turn result

Disc record. Disc record: the quarter view reveals a flat circular face. Ask which model has one flat circular face.

Edge result

Disc record. Disc record: its edge-on outline is very thin. The sphere has no flat edge direction that becomes a line.

Front result

Sphere model record. Sphere record: retain this result and compare it with the other views. Use the label set: this card belongs to the object that remains round across all three results.

Quarter-turn result

Sphere model record. Sphere record: a curved surface keeps the outline round. The disc becomes oval at this viewpoint.

Edge-direction result

Sphere model record. Sphere record: there is no thin flat edge to expose. Choose the object that stays round rather than narrowing to a line.

Changing spacecraft views

Real space record. Real space record: many changing views show rounded whole bodies with no thin flat edge. This card reports views of the real Earth and Moon, not a classroom model.

Earth's eclipse shadow

Real space record. Real space record: the curved shadow agrees with an approximately spherical Earth. This is a real sky observation of Earth's shadow.

Moving day-night edge

Real space record. Real space record: the curved moving light boundary fits rounded three-dimensional bodies. Sort this as real evidence seen in space images.

Sun's curved edge

Real space record. Real space record: repeated Sun observations agree with an approximately spherical body. This card describes repeated observations of the real Sun.

Model limitation

The models show overall shape but not real size, surface detail, gravity, distance or the slight flattening of rotating bodies.

Independent evidence

Six model observations, one named real space observation, an approximately spherical conclusion and one model limitation.

Supported: Complete a six-row picture table and choose the conclusion supported. Use outline choices: round, oval, line. Finish: This supports __ because....

Standard: Record all six model observations, add one real space observation and justify the approximately spherical description. Cite a matched model view and a real observation; keep observations separate from the conclusion.

Stretch: Evaluate the evidence, identify an alternative explanation from one view and state why combined views are stronger. Include a fair comparison, several pieces of evidence that agree and a model limitation.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: [SCI_G_W9B_Spherical_Bodies_Do.html](#)

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.